NSW GOVERNMENT

NSW Education Standards Authority

Centre Number

Student Number

2026 **HIGHER SCHOOL CERTIFICATE EXAMINATION**

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page

Total marks: 100

Section I – 10 marks (pages 2–9)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 13–44)

- Attempt Questions 11–33
- Allow about 2 hours and 45 minutes for this section

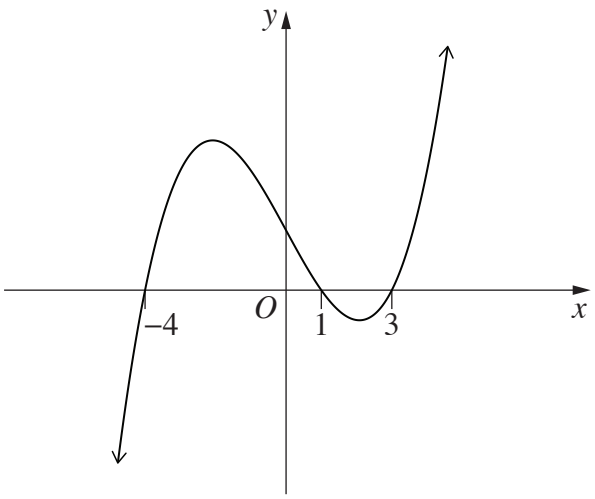
Section I

10 marks
Attempt Questions 1–10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

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1 A graph of $y = f(x)$ is shown.



Which of the following expressions best represents $f(x)$?

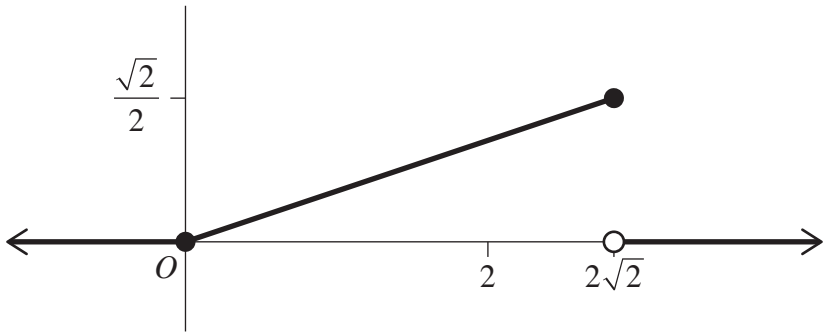
A. $-(x-1)(x-3)(x+4)$

B. $-(x+1)(x+3)(x-4)$

C. $(x-1)(x-3)(x+4)$

D. $(x+1)(x+3)(x-4)$

2 A probability density function is shown in the diagram.



What is the mode of the given probability density function?

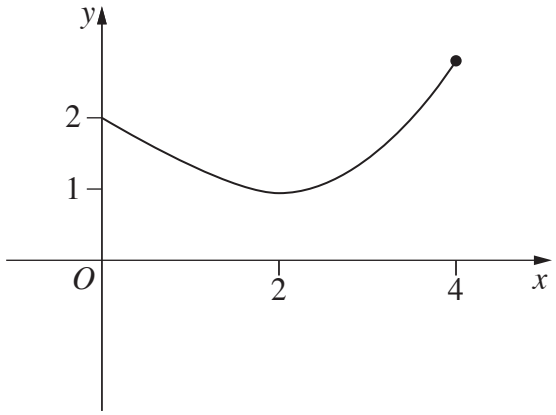
A. 0

B. $\frac{\sqrt{2}}{2}$

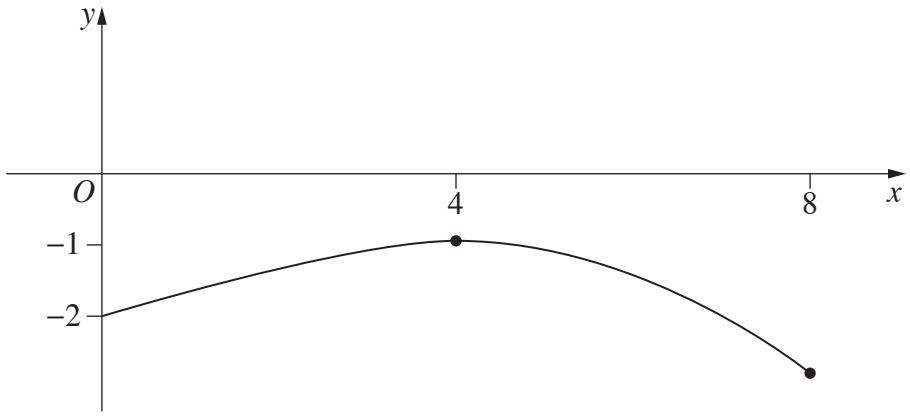
C. 2

D. $2\sqrt{2}$

3 The graph of $y = f(x)$ is shown for $0 \leq x \leq 4$.



A new graph $y = g(x)$ is obtained by transforming $y = f(x)$ as shown.



Which of the following expressions is equivalent to $g(x)$?

A. $f\left(-\frac{1}{2}x\right)$

B. $-f\left(\frac{1}{2}x\right)$

C. $f(-2x)$

D. $-f(2x)$

4 A study of Year 8 students shows that there is a positive correlation between the number of hours spent studying each week and the students' test scores. The Pearson's correlation coefficient is 0.62.

From this information it can be concluded that

A. 62% of students who study more hours achieve higher test scores.

B. the number of hours students study causes an increase in test scores.

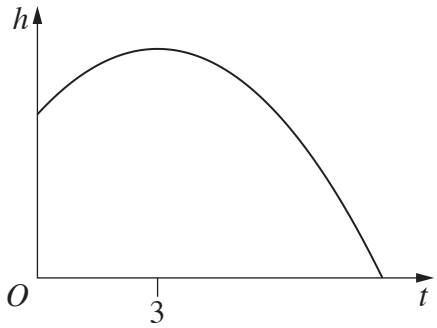
C. 62% of students with higher test scores choose to study more hours each week.

D. students who study for more hours each week tend to achieve higher test scores.

5 An object is thrown vertically upwards from the top of a tall building. The height, h metres, of the object above the ground after t seconds is given by

$$h = 45 + 12t - 2t^2.$$

The object reaches its maximum height 3 seconds after it is thrown. The diagram shows the graph of $h(t)$ for $t \geq 0$.



What is the vertical distance, in metres, that the object rises above the top of the building?

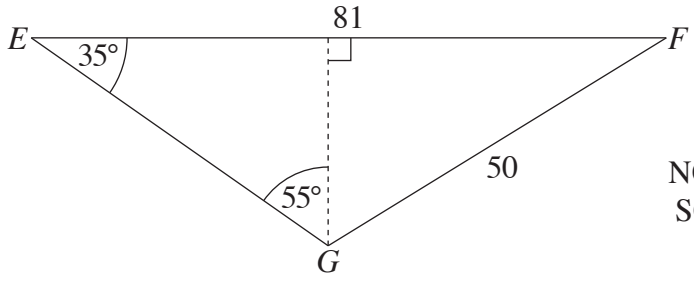
A. 18

B. 27

C. 45

D. 63

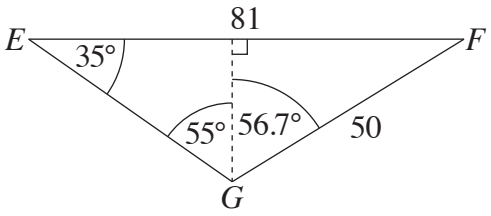
6 A triangle is shown.



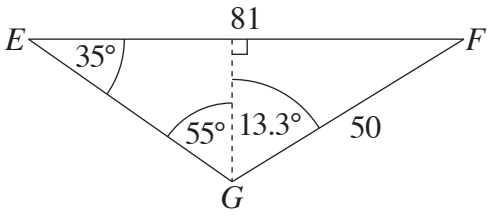
It is given that $\sin^{-1}\left(\frac{81 \times \sin 35°}{50}\right) = 68.3°$, correct to 1 decimal place.

Which diagram shows the correct angle sizes at $\angle EGF$, given that the angle is obtuse?

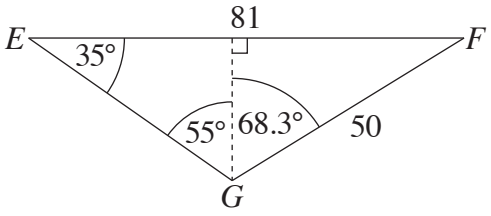
A.



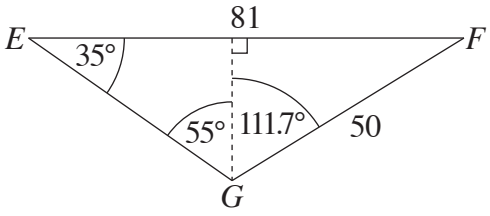
B.



C.



D.



7 A biased six-sided die has faces numbered $-4, -3, 0, 1, 2$ and 6.

The table shows the probability distribution for X, the number which lands face up after the die is rolled.

X	-4	-3	0	1	2	6
$P(X)$	$\frac{3}{18}$	$\frac{4}{18}$	$\frac{3}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{6}{18}$

The die is rolled twice.

What is the probability that the sum of the two numbers rolled is -3?

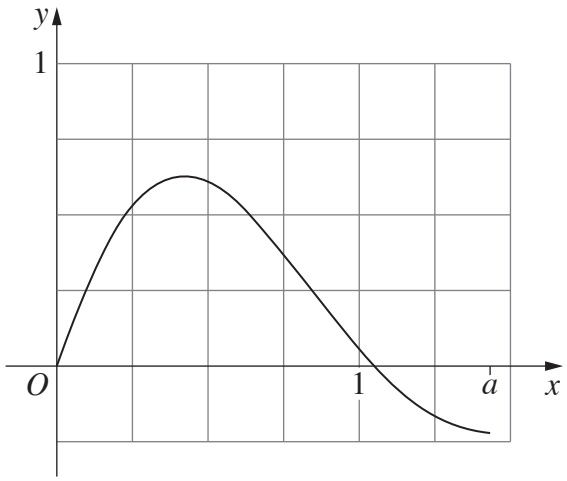
A. $\frac{1}{54}$

B. $\frac{5}{54}$

C. $\frac{4}{18}$

D. $\frac{6}{18}$

8 A section of the graph of $y = f(x)$ for $x \geq 0$ is shown, where $f(x)$ is an odd function.



Four integrals based on this graph are given.

$$P = \int_0^a f(x)\,dx \qquad R = \int_{-a}^1 f(x)\,dx$$

$$Q = \int_{-a}^a f(x)\,dx \qquad S = \int_{-a}^0 f(x)\,dx$$

Which of the following options identifies the integrals in ascending numerical order?

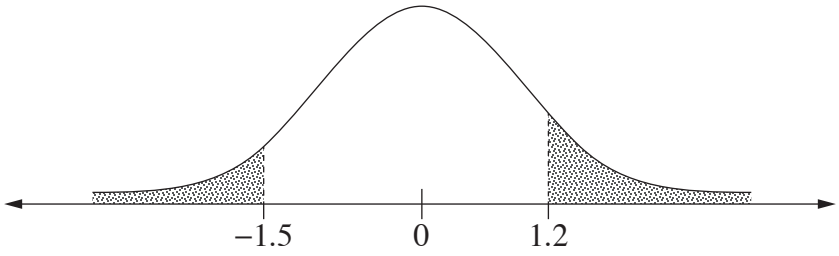
A. P, R, S, Q

B. P, S, R, Q

C. S, Q, R, P

D. S, R, Q, P

9 The regions containing standardised scores less than -1.5 and greater than 1.2 have been shaded on the normal distribution shown.



Which expression gives the probability that a score selected at random falls in the shaded regions?

A. $2P(Z < -1.5) + P(-1.5 < Z < -1.2)$

B. $2P(Z < -1.2) + P(1.2 < Z < 1.5)$

C. $2P(Z < 1.2) - P(-1.5 < Z < 1.2)$

D. $2P(Z < 1.5) - P(1.2 < Z < 1.5)$

10 It is given that $h(x) = \sin^2 x - \sin x$ is a composite function of $f(x)$ and $g(x)$ where $f(x) = x^2 - x$ and $g(x) = \sin x$ for all real values of x.

For what values of k does the equation $\sin^2 x - \sin x = k$ have real solutions?

A. $-1 \le k \le 1$

B. $-1 \le k \le 2$

C. $-\frac{1}{4} \le k \le 1$

D. $-\frac{1}{4} \le k \le 2$

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2026 HIGHER SCHOOL CERTIFICATE EXAMINATION

Centre Number

Student Number

Student Name

Mathematics Advanced

Section II Answer Booklet

90 marks
Attempt Questions 11–33
Allow about 2 hours and 45 minutes for this section

Instructions

- Write your Centre Number, Student Number and Student Name at the top of this page
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response
- Your responses should include relevant mathematical reasoning and/or calculations
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering

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Question 11 (2 marks)

Find the gradient of the tangent to $y = x^3 + 3x$ at $x = -1$. **2**

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Question 12 (4 marks)

The number of cars parked in a carpark is recorded each day for thirteen days. The dataset is listed in ascending order:

12, 15, 15, 17, 19, 20, 22, 24, 25, 27, 28, 30 and m.

(a) What is the interquartile range of this dataset? **2**

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(b) What is the greatest value of m such that this dataset has no outlier? **2**

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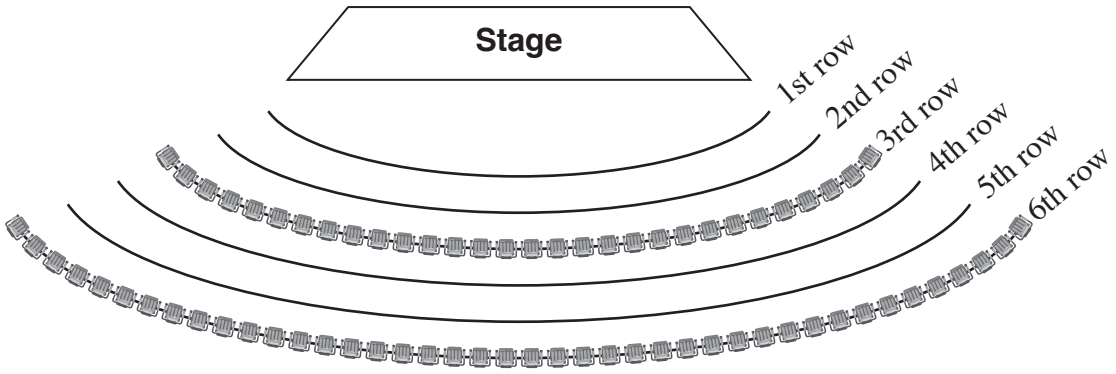
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Question 13 (4 marks)

In a theatre, starting from the stage, the number of seats in each row forms an arithmetic sequence.

The theatre has 30 seats in the 3rd row and 42 seats in the 6th row.



(a) How many seats are in the first row? **2**

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(b) How many seats will this theatre have if there are 9 rows? **2**

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Question 14 (2 marks)

Find the range and period of the function $h(x) = 3\cos(2x) + 1$. **2**

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Question 15 (2 marks)

Find $\frac{dy}{dx}$ if $y = 4^{x^2+3}$. **2**

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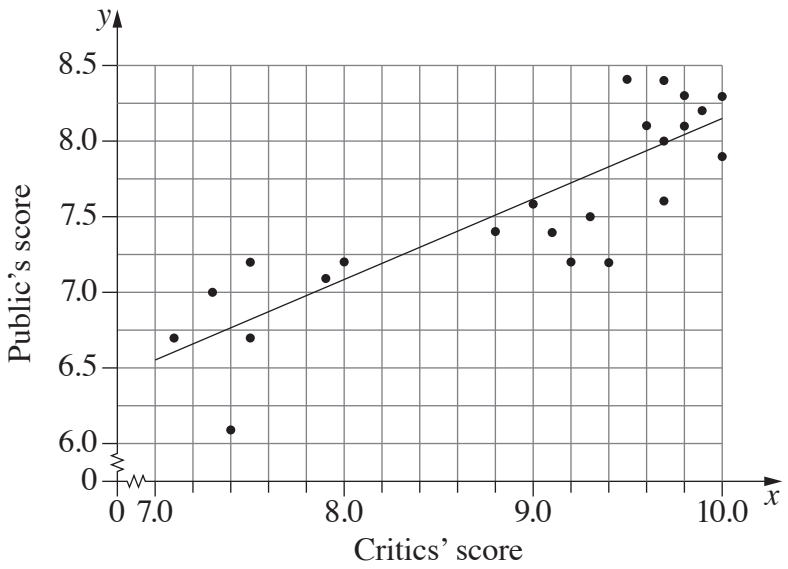
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Question 16 (4 marks)

Researchers were interested in whether the critics' scores for movies could be used to predict the general public's scores for movies.

A random sample of movies was selected, and the average score from the critics and the average score from the public were recorded.

A scatterplot of the dataset and the least-squares regression line are shown.



(a) What was the lowest score given by the public for a movie, correct to 1 decimal place? **1**

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(b) The gradient of the least-squares regression line for the dataset is 0.54. **2**

Interpret the gradient in the context of the dataset.

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(c) A new data point is included in the dataset. The critics' score is 7.2 and the public's score is 6. **1**

Identify ONE effect this would have on the least-squares regression line.

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Question 17 (3 marks)

A large rechargeable battery has a maximum storage capacity of 40 kWh when new.

After 1 year, the maximum storage capacity is 32 kWh.

The process of decay can be modelled by the function

$$y = ka^t$$

where y is the maximum storage capacity of the battery, in kWh, t years after it starts being used.

(a) What are the values of k and a? **2**

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(b) Interpret the value of a in this context. **1**

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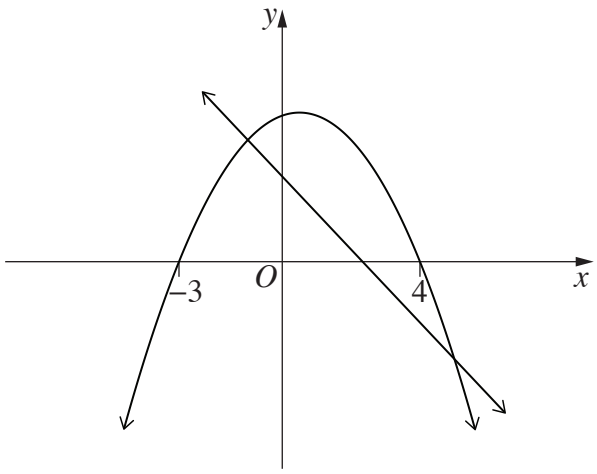
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Question 18 (4 marks)

The diagram shows the graphs of $y = 7 - 3x$ and $y = -x^2 + x + 12$. The graphs intersect at two points.



(a) Show that the x-coordinates of the two points of intersection are $x = -1$ and $x = 5$. **2**

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(b) Hence find the area enclosed by the line $y = 7 - 3x$ and the parabola $y = -x^2 + x + 12$. **2**

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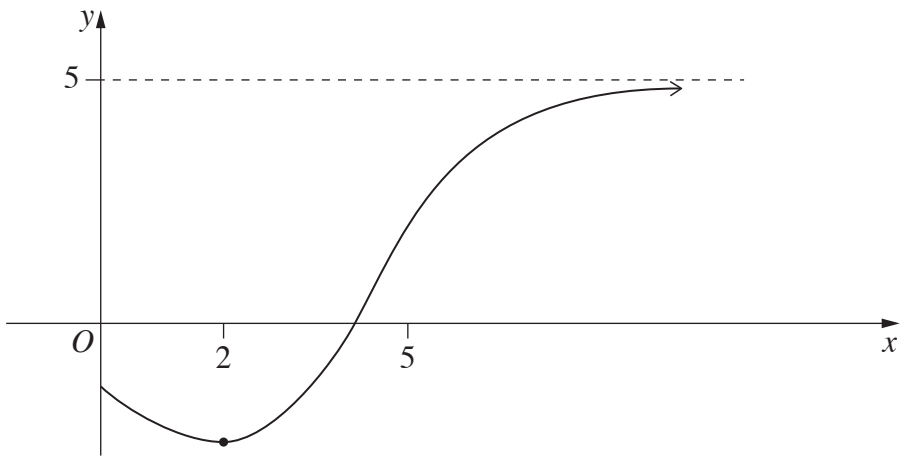
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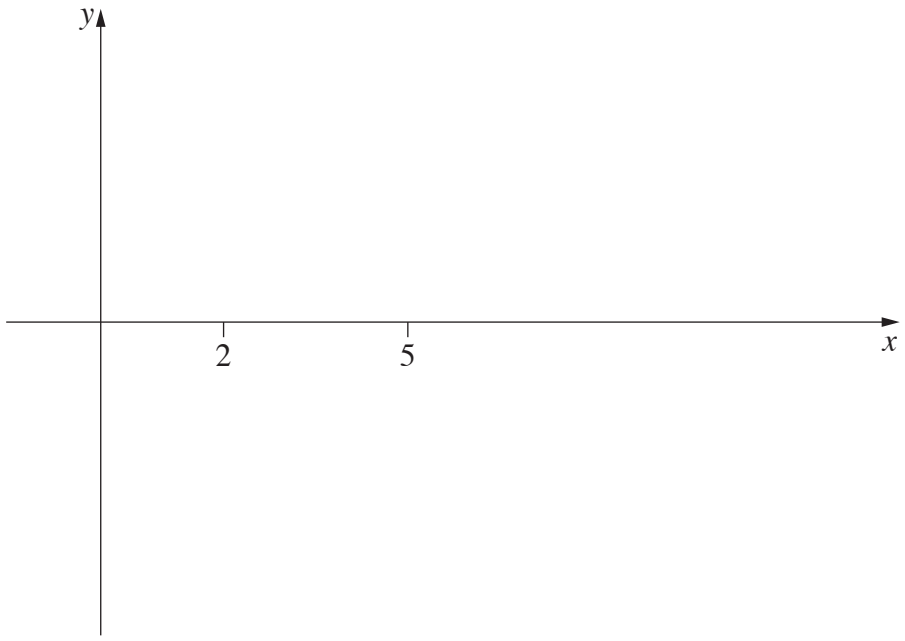
Question 19 (3 marks)

A graph of $y = f(x)$ is shown. The graph of $f(x)$ has a local minimum at $x = 2$ and a point of inflection at $x = 5$. **3**

The function also has a horizontal asymptote at $y = 5$.



On the axes provided, sketch the graph of $y = f'(x)$.



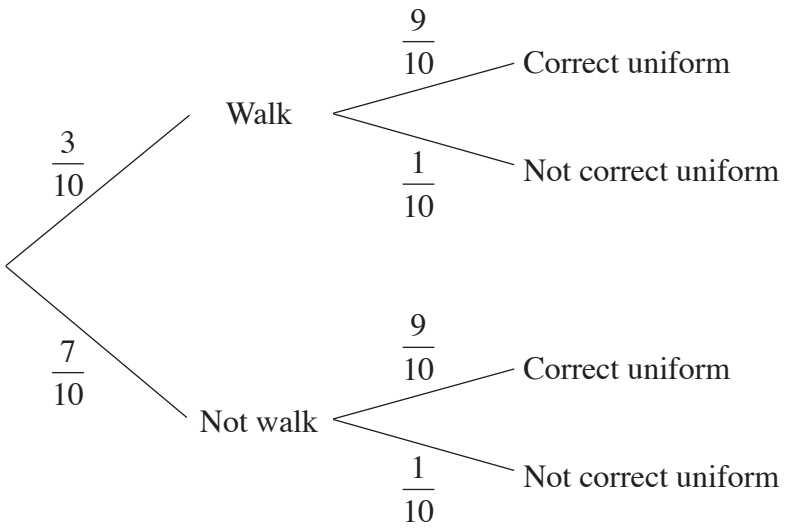
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Question 20 (3 marks)

At a particular school, the probability that a student walks to school is $\frac{3}{10}$ and the probability that a student wears the correct school uniform is $\frac{9}{10}$.

The probability tree shows this information.



(a) One student at the school is selected at random. **1**

What is the probability that this student walks to school and is not wearing the correct school uniform?

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(b) Two students at the school are selected at random. **2**

What is the probability that at least one of these students walks to school and is not wearing the correct school uniform?

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Question 21 (5 marks)

(a) Show that the first derivative of $g(x) = \dfrac{x}{\ln x}$ is $g'(x) = \dfrac{\ln x - 1}{(\ln x)^2}$. **2**

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(b) What is the global maximum of $g(x)$ on the interval $[2, 5]$? **3**

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Question 22 (3 marks)

(a) Show that $\frac{x+2}{x+1} = 1 + \frac{1}{x+1}$. **1**

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(b) Describe the transformations required to obtain the graph $y = \frac{x+2}{x+1}$ from the graph of $y = \frac{1}{x}$. **2**

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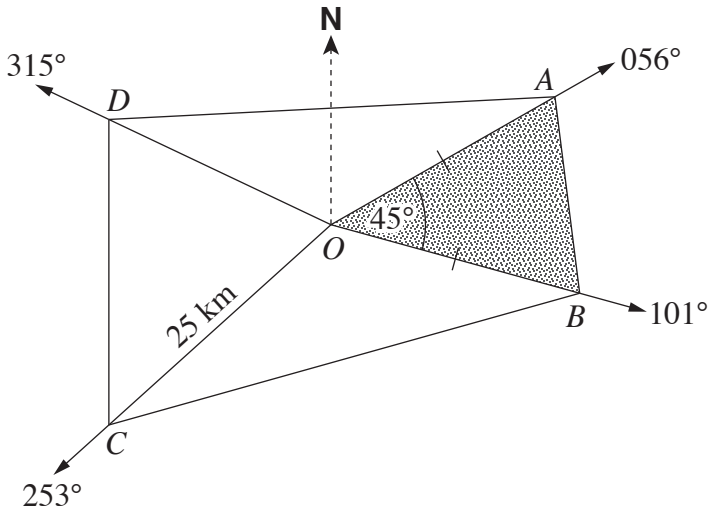
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Question 23 (4 marks)

The diagram shows a block of land *ABCD*. It also shows the bearing of each of the points *A*, *B*, *C*, *D*, from point *O*. **4**



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The distance from *O* to *C* is 25 km, $OA = OB$ and $\angle AOB = 45°$.

The area of the shaded triangle *AOB* is 80 km^2.

What is the distance from *B* to *C*, to the nearest kilometre?

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Question 24 (4 marks)

Radioactive cobalt-60 decays to become nickel-60. A 1000 g block of radioactive cobalt-60 is placed in a container.

After a period of 10.54 years, 75% of the cobalt-60 has decayed to become nickel-60.

The decay process can be modelled by the exponential relationship

$$A = 1000e^{kt}$$

where A is the amount of cobalt-60 remaining (in grams) at time t (in years).

(a) Find the value of k, correct to 4 decimal places. **2**

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(b) Find the value of t when the rate of change of A is -9.48 g/year, correct to the nearest year. **2**

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Question 25 (2 marks)

The weights of watermelons produced by a certain farm are normally distributed with a mean of 4.82 kg. Long-term quality control data show that when a watermelon from this farm is randomly selected, the probability that it weighs less than 3.75 kg is 0.116. **2**

A watermelon is labelled as being the ideal size if it weighs between 4.82 kg and 5.89 kg.

What is the probability that a watermelon selected at random is the ideal size?

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Questions 11–25 are worth 49 marks in total

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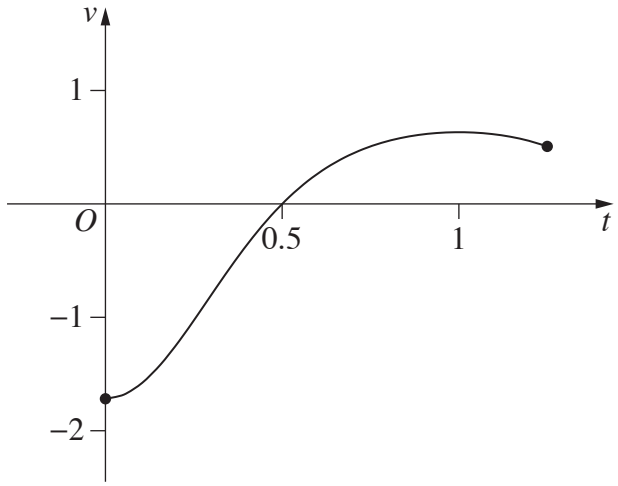
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Question 26 (5 marks)

The velocity, v m/s, of a particle moving in a straight line is given by the function

$$v(t) = 1 - e^{\cos(\pi t)}$$

where $0 \leq t \leq 1.25$ is the time in seconds. The graph of $v(t)$ is shown.



(a) At what value(s) of t does the particle change direction in this interval? **1**

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(b) Find the acceleration of the particle when $t = 0.5$. **2**

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Question 26 continues on page 29

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Question 26 (continued)

(c) The table provides five function values for v, correct to 2 decimal places. **2**

t	0	0.25	0.5	0.75	1
v	−1.72	−1.03	0	0.51	0.63

Using the trapezoidal rule with four subintervals, find the approximate distance travelled by the particle in the interval $0 \le t \le 1$.

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End of Question 26

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Question 27 (4 marks)

Maya plans to go on an overseas trip in 5 years' time. Maya expects that the trip will cost $14 000. To save for the trip, Maya began by contributing $750 every 6 months into an annuity. After making a number of payments, Maya realised that the annuity would not reach $14 000 in 5 years. Maya decides to double the contribution until the trip. **4**

The interest rate was 2% per annum compounding half yearly.

Future values of an annuity of $1

Rate / *Period*	*0.01*	*0.02*	*0.03*	*0.04*
2	2.0100	2.0200	2.0300	2.0400
4	4.0604	4.1216	4.1836	4.2465
6	6.1520	6.3081	6.4684	6.6330
8	8.2857	8.5830	8.8923	9.2142
10	10.4622	10.9497	11.4639	12.0061
12	12.6825	13.4121	14.1920	15.0258

What is the maximum number of $750 contributions Maya could make before she will need to double her payments to reach $14 000?

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Question 28 (4 marks)

Let $y = 4x^3 - 3x^2 - 6x + d$, where d is a real number. **4**

Use differential calculus to find the values of d for which this curve has exactly three x-intercepts. (Note: You do NOT need to find the x-intercepts.)

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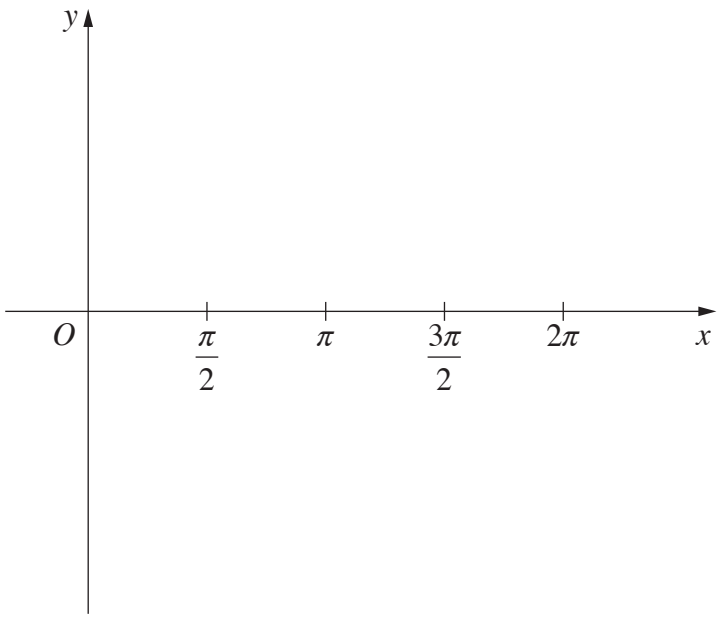
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Question 29 (6 marks)

(a) Sketch the graph of $y = \cot x$ on the axes provided for $0 \leq x \leq 2\pi$. **2**



(b) Hence, or otherwise, find the values of x for which $y = \ln(\cot x)$ is defined in the interval $0 \leq x \leq 2\pi$. **2**

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Question 29 continues on page 33

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Question 29 (continued)

(c) Prove that $\frac{d}{dx}\left(\ln(\cot x)\right) = -\operatorname{cosec} x \sec x$. **2**

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End of Question 29

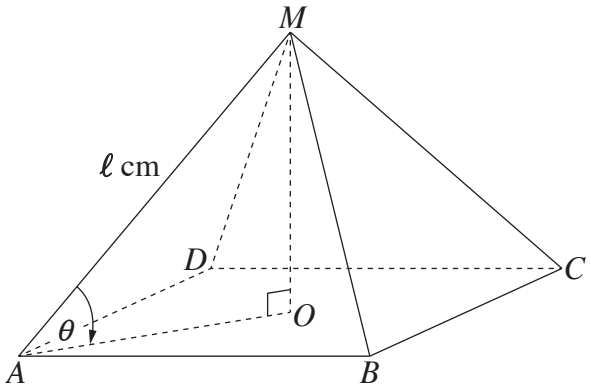
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Question 30 (7 marks)

A pyramid $MABCD$ has its vertex M vertically above O, the centre of its square base $ABCD$, as shown. $AM = \ell$ cm and makes an angle of θ with the base.



(a) Let V be the volume of the pyramid. **3**

Show that $V = \frac{2\ell^3}{3} \sin\theta \cos^2\theta$.

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Question 30 (continued)

(b) Hence, find the value of θ, in radians, for which the volume of the pyramid is a maximum. Give your answer correct to 2 decimal places. **4**

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Question 31 (4 marks)

(a) Find $\int \cos 2x \, dx$. **1**

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(b) It is given **3**

$$\int \frac{f'(x)}{f(x)} dx = \int \cos 2x \, dx$$

where $f(0) = 1$ and $f(x) > 0$ for all real x.

Find $f(x)$.

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Question 32 (3 marks)

Let

$$g(x) = \int_{x^3}^{1} \sqrt{\tan(t)}\, dt$$

and let $F(t)$ be an antiderivative of $\sqrt{\tan(t)}$, where $0 \le x \le 1$. (Do NOT try to find $F(t)$.)

(a) Find an expression for $g(x)$ in terms of F. **1**

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(b) Hence find $g'(x)$. **2**

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Question 33 (8 marks)

Peyton borrows $83 000. Any amount owing on the loan attracts interest at an annual rate of 6%, compounded monthly.

The repayments are made at the end of each month after the interest is added to the account.

Peyton makes a first monthly repayment of $760 and then increases the value of the repayment by 10% each month.

Let A_n = Amount owing after the nth repayment.

(a) Show that $A_3 = 83\,000(1.005)^3 - 760\big((1.005)^2 + (1.005)(1.1) + (1.1)^2\big)$. **2**

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(b) Show that $A_n = 91\,000(1.005)^n - 8000(1.1)^n$. **3**

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Question 33 (continued)

(c) Find the amount of the final payment. **3**

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2026 **HIGHER SCHOOL CERTIFICATE EXAMINATION**

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}\left[2a + (n - 1)d\right] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, \; r \neq 1$$

$$S = \frac{a}{1 - r}, \; |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

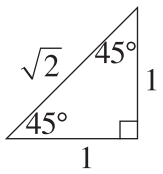
$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

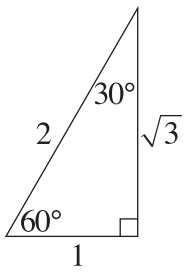
$$A = \frac{1}{2}ab\sin C$$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$

Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \ \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \ \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \ \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A\cos B + \cos A\sin B$$

$$\cos(A + B) = \cos A\cos B - \sin A\sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

If $t = \tan\frac{A}{2}$ then $\sin A = \frac{2t}{1 + t^2}$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A\cos B = \frac{1}{2}\left[\cos(A - B) + \cos(A + B)\right]$$

$$\sin A\sin B = \frac{1}{2}\left[\cos(A - B) - \cos(A + B)\right]$$

$$\sin A\cos B = \frac{1}{2}\left[\sin(A + B) + \sin(A - B)\right]$$

$$\cos A\sin B = \frac{1}{2}\left[\sin(A + B) - \sin(A - B)\right]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

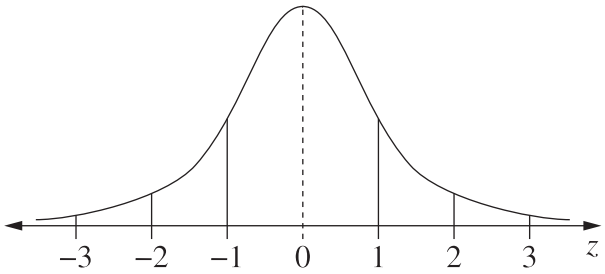
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between –1 and 1
- approximately 95% of scores have z-scores between –2 and 2
- approximately 99.7% of scores have z-scores between –3 and 3

$$E(X) = \mu$$

$$\text{Var}(X) = E\left[(X - \mu)^2\right] = E\left(X^2\right) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x)\,dx$$

$$P(a < X < b) = \int_a^b f(x)\,dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r(1 - p)^{n-r}$$

$$X \sim \text{Bin}(n, p)$$

$$\Rightarrow \ P(X = x)$$

$$= \binom{n}{x} p^x(1 - p)^{n-x}, \ x = 0, 1, \ldots, n$$

$$E(X) = np$$

$$\text{Var}(X) = np(1 - p)$$

Differential Calculus

Function	Derivative
$y = f(x)^n$	$\frac{dy}{dx} = nf'(x)\left[f(x)\right]^{n-1}$
$y = uv$	$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$
$y = g(u)$ where $u = f(x)$	$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
$y = \frac{u}{v}$	$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
$y = \sin f(x)$	$\frac{dy}{dx} = f'(x)\cos f(x)$
$y = \cos f(x)$	$\frac{dy}{dx} = -f'(x)\sin f(x)$
$y = \tan f(x)$	$\frac{dy}{dx} = f'(x)\sec^2 f(x)$
$y = e^{f(x)}$	$\frac{dy}{dx} = f'(x)e^{f(x)}$
$y = \ln f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\frac{dy}{dx} = (\ln a) f'(x)a^{f(x)}$
$y = \log_a f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$
$y = \sin^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - \left[f(x)\right]^2}}$
$y = \cos^{-1} f(x)$	$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - \left[f(x)\right]^2}}$
$y = \tan^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{1 + \left[f(x)\right]^2}$

Integral Calculus

$$\int f'(x)\left[f(x)\right]^n dx = \frac{1}{n+1}\left[f(x)\right]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x)\sin f(x)\,dx = -\cos f(x) + c$$

$$\int f'(x)\cos f(x)\,dx = \sin f(x) + c$$

$$\int f'(x)\sec^2 f(x)\,dx = \tan f(x) + c$$

$$\int f'(x)e^{f(x)}dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)}dx = \ln\left|f(x)\right| + c$$

$$\int f'(x)a^{f(x)}dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - \left[f(x)\right]^2}}dx = \sin^{-1}\frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + \left[f(x)\right]^2}dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a} + c$$

$$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$$

$$\int_a^b f(x)\,dx$$

$$\approx \frac{b-a}{2n}\left\{f(a) + f(b) + 2\left[f\left(x_1\right) + \cdots + f\left(x_{n-1}\right)\right]\right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^{n}P_{r} = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^{n}C_{r} = \frac{n!}{r!(n-r)!}$$

$$(x+a)^{n} = x^{n} + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^{r} + \cdots + a^{n}$$

Vectors

$$\left| \underset{\sim}{u} \right| = \left| x\underset{\sim}{i} + y\underset{\sim}{j} \right| = \sqrt{x^{2} + y^{2}}$$

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = \left| \underset{\sim}{u} \right| \left| \underset{\sim}{v} \right| \cos\theta = x_{1}x_{2} + y_{1}y_{2},$$

where $\underset{\sim}{u} = x_{1}\underset{\sim}{i} + y_{1}\underset{\sim}{j}$

and $\underset{\sim}{v} = x_{2}\underset{\sim}{i} + y_{2}\underset{\sim}{j}$

$$\underset{\sim}{r} = \underset{\sim}{a} + \lambda\underset{\sim}{b}$$

Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta)$$
$$= re^{i\theta}$$

$$\left[r(\cos\theta + i\sin\theta)\right]^{n} = r^{n}(\cos n\theta + i\sin n\theta)$$
$$= r^{n}e^{in\theta}$$

Mechanics

$$\frac{d^{2}x}{dt^{2}} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d}{dx}\left(\frac{1}{2}v^{2}\right)$$

$$x = a\cos(nt + \alpha) + c$$

$$x = a\sin(nt + \alpha) + c$$

$$\ddot{x} = -n^{2}(x - c)$$

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